

Physical Activity & BMI Percentile

Research Question

Does physical activity affect BMI percentile among high school students?

Method

- Descriptive Statistics
- Correlation Analysis
- Linear Regression
- Gender Comparison Analysis (Extended)

Variable

Exercise Days(0–7) Independent
Gender (Male/Female) Independent
BMI Percentile(%) Dependent

Interpretation

- Students who exercised more frequently tended to have slightly lower BMI percentile values
- However, the relationship was extremely weak and not statistically significant
- Gender-based analysis showed slight differences in BMI percentile distributions between male and female students
- Gender may influence the relationship between physical activity and BMI percentile

Key Result

$n=12,527$ $r=-0.013$ Slope= -0.14
 $R^2=0.0002$ P-value= 0.145

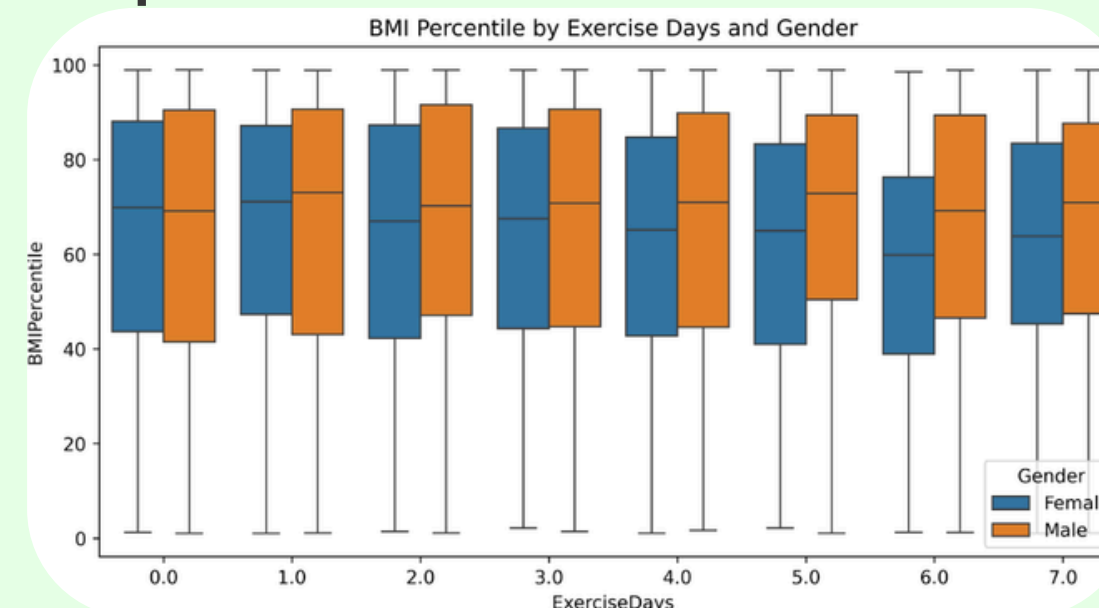
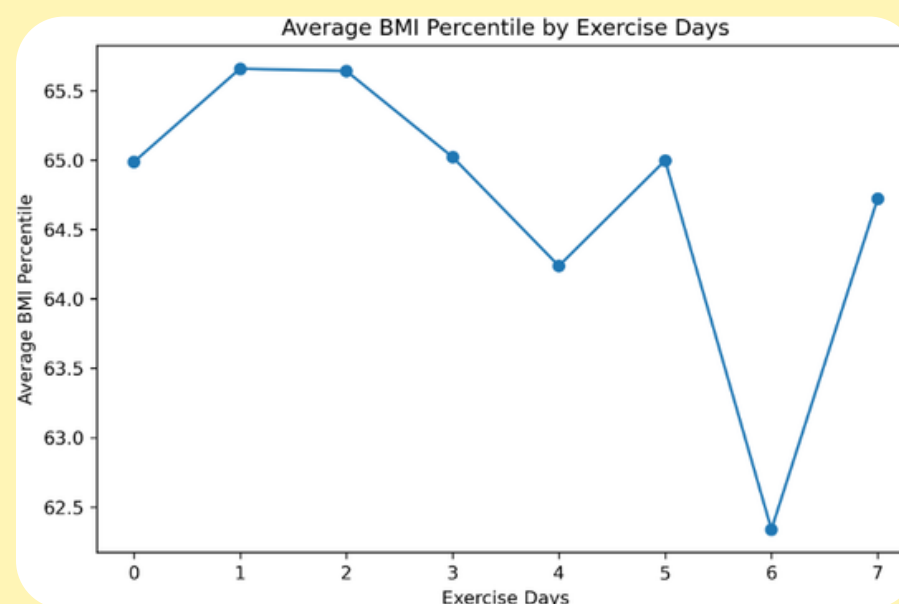
Regression Equation
 $BMI_{Percentile}=65.2935-0.1393 \times ExerciseDays$

Exercise frequency explains almost none of the variation in BMI percentile

Conclusion

- No significant relationship was found between physical activity and BMI percentile.
- Small gender differences were observed in BMI percentile distributions.
- Gender may influence the relationship between exercise and BMI.

Exercise & BMI



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- Gender may moderate the relationship between exercise and BMI.